



Technical, Allocative, and Economic Efficiency of Organic Rice Farming in Magelang Regency, Central Java, Indonesia

Eni Istiyanti^{1,2*}, Ashari², Pujiati Utami³, Alim Fausul Rouf¹

¹ Department of Agribusiness, Universitas Muhammadiyah Yogyakarta, 55183 Yogyakarta, Indonesia

² Research Center for Behavioral and Circular Economic, National Research and Innovation Agency (BRIN), 12710 Jakarta Selatan, Indonesia

³ Department of Agribusiness, Universitas Muhammadiyah Puwokerto, 53182 Banyumas, Indonesia

* Correspondence: Eni Istiyanti (eniistiyanti@umy.ac.id)

Received: 11-29-2025

Revised: 02-26-2026

Accepted: 03-05-2026

Citation: Istiyanti, E., Ashari, Utami, P., & Rouf, A. F. (2026). Technical, allocative, and economic efficiency of organic rice farming in Magelang Regency, Central Java, Indonesia. *Org. Farming*, 12(2), 73–85. <https://doi.org/10.56578/of120201>.



© 2026 by the author(s). Published by Acadlore Publishing Services Limited, Hong Kong. This article is available for free download and can be reused and cited, provided that the original published version is credited, under the CC BY 4.0 license.

Abstract: Magelang Regency has been designated as a pilot area for organic rice development, supported by an integrated institutional framework that includes farmer training, extension services, and infrastructure development. This study analyzed the technical efficiency (TE), allocative efficiency (AE), and economic efficiency (EE) of organic rice farming in Magelang Regency, within the context of a government-supported organic farming development program. A stochastic frontier analysis (SFA) was applied to cross-sectional data at farm level to estimate production efficiency and to examine the role of socioeconomic factors in explaining inefficiency. The results indicated relatively high average TE among organic rice farmers. However, estimated inefficiency components were small and statistically insignificant, suggesting that variations in output were largely influenced by stochastic and exogenous factors rather than by technical inefficiency. AE was found to be more moderate, leading to lower EE despite favorable technical performance. This pattern implies that considerations of cost allocation and input price remain important constraints in organic rice farming systems. Socioeconomic characteristics such as age, education, farming experience, and participation of farmer group were associated with lower estimated inefficiency. The findings contribute expectedly to the efficiency of literature on organic agriculture, while offering policy-relevant evidence for improving the design and implementation of organic rice development programs.

Keywords: Allocative efficiency; Organic rice; Stochastic frontier analysis; Technical inefficiency

1. Introduction

Organic farming is an agricultural system that utilizes local resources (Piadozo et al., 2014). This system aims not only to improve the quality of agricultural products but also to maintain the balance of the ecosystem and human health (Pretty & Bharucha, 2014). The implementation of organic farming could reduce the dependence of farmers on chemical-based agricultural inputs, such as synthetic fertilizers and pesticides, which in the long term can damage soil fertility and cause environmental pollution (Reganold & Wachter, 2016). Organic farming system has advantages over conventional farming in terms of social, economic, and environmental aspects (Phantha et al., 2021).

Rice (*Oryza sativa* L.), as a major food commodity in Indonesia, has great potential for the development of organic farming. The cultivation of organic rice focuses on environmentally-friendly methods that do not use chemical fertilizers or synthetic pesticides. In addition to producing healthy food, this cultivation also maintains soil fertility and supports environmental sustainability. This makes organic rice a strategic choice to support a sustainable agricultural system (David & Ardiansyah, 2020).

Market demand for organic products, including organic rice, continues to increase both domestically and internationally (Willer et al., 2021). This trend provides a great opportunity for farmers to switch to a more environmentally-friendly organic farming system. In addition, practices of organic farming enhance food security

by producing healthier products that are free of harmful chemical residues (Willer & Lernoud, 2019).

Despite these opportunities, organic rice farming faces substantial challenges, particularly during the transition from conventional systems. Previous studies have reported yield declines, higher labor requirements, market uncertainty, and certification costs as major constraints affecting farmers' performance and willingness to adopt organic practices (Nair et al., 2014; Sujianto et al., 2024). These constraints directly influence production efficiency and, ultimately, the economic sustainability of organic rice farming.

A substantial body of literature has examined the technical efficiency (TE), allocative efficiency (AE), and economic efficiency (EE) of organic rice farming in Indonesia and other countries using stochastic frontier analysis (SFA) and related approaches. However, most of this research treated relatively homogeneous organic rice farms and was limited to specific institutional arrangements, policy interventions, and regional pilot programs that allowed efficient outcomes (Hidalgo et al., 2025; Hidayati et al., 2019; Istiyanti & Rahmadynda, 2025; Panpluem et al., 2019). Empirical evidence on how large-scale and government-supported organic rice development programs affect farmer efficiency remains limited.

Magelang Regency, Central Java, represents a particularly informative case. The region has been designated as a pilot area for organic rice development, supported by an integrated institutional framework that includes farmer training, extension services, and infrastructure development. From 2021 to 2024, Magelang has been a key beneficiary of the UPLAND Program launched by the Ministry of Agriculture, which targets the development of organic rice farming across approximately 2,000 hectares of highland areas (International Fund for Agricultural Development, 2019; International Fund for Agricultural Development, 2025). In many other organic rice-producing regions, however, adoption is largely farmer-initiated and less institutionally coordinated. The combination of policy support, institutional coordination, and specific agroecological conditions clearly differentiates Magelang from other organic rice-producing regions in Indonesia.

This study explicitly focused on assessing the TE, AE, and EE of organic rice farming in Magelang Regency, Central Java, Indonesia under an institutionalized development framework, rather than under generic or purely farmer-driven organic adoption contexts. By doing so, this research provides new insights into the extent to which government-supported organic farming initiatives translate into measurable efficiency outcomes at the farm level. The findings expectedly contribute to the efficiency literature on organic agriculture, while offering policy-relevant evidence for improving the design and implementation of organic rice development programs in Indonesia and other developing countries.

2. Methodology

2.1 Research Location

This research was conducted in Magelang Regency, as shown in Figure 1. This location was chosen because Magelang Regency is the center of organic rice production in Indonesia. In addition, this location has joined the UPLAND project of the government and has obtained organic certification LeSOS (Lembaga Sertifikasi Organik Seloliman) for its farming groups.

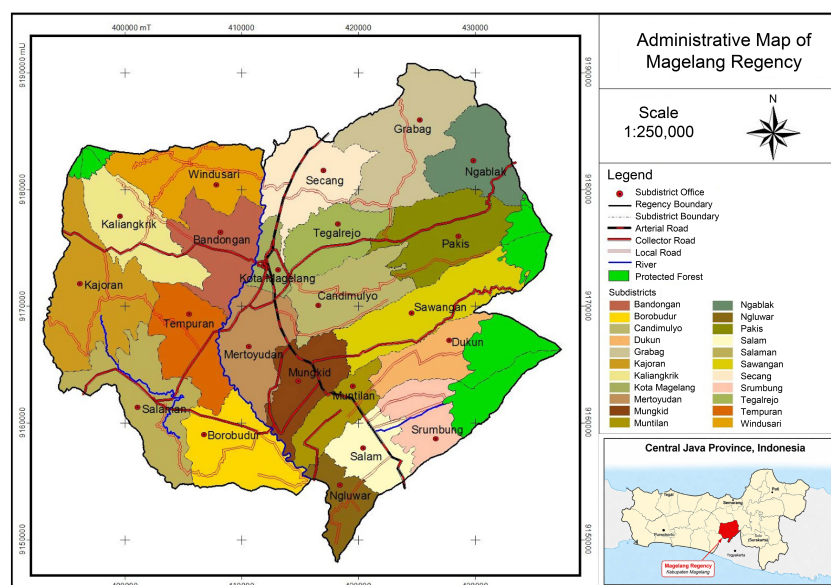


Figure 1. Research location
Source: Adapted from Pemerintah Kabupaten Magelang (2018)

2.2 Sample Procedures and Data Collection

The informants in this study were selected using a proportional random sampling method, with a total of 125 respondents from 22 organic rice farmer groups in Magelang Regency. In the study area, there are 34 farmer groups whose members have been practicing organic rice farming since 2012. Among these, 22 groups cultivate organic rice every season without intercropping, while 12 groups grow horticultural crops in certain seasons alongside rice. Data collection was conducted through questionnaire-guided structured interviews. The data used in this study were obtained from the planting season from January to April 2025 in Magelang Regency. The research samples are presented in Table 1.

Table 1. Number of research respondents

Farmer Group	Number of Respondents (Farmers)
Sedyo Raharjo	5
Sido Rukun	6
Ngudi Rejeki	3
Sedyo Utomo	6
Ngudi Makmur	4
Permata Tani	6
Kebo Kuning	2
Pasti Mulyo	15
Margorejo	9
Tirto Semaren	6
Sumber Rejeki	5
Maju Makmur	7
Makmur Barokah	2
Karya Tani	3
Rukun Tani	7
Tri Rejeki	4
Ngudi Makmur Keron	5
Suko Tulodo	7
Denokan	3
Piyungan Barat	3
Piyungan Tengah	6
Ngudi Rejeki	11
Total	125

2.3 Techniques of Analysis

The data analysis method used in this study was the SFA with the Cobb Douglass function approach (Coelli et al., 2005). The advantages of SFA are its ability to specifically define the form of cost, profit, or production functions, estimate the level of efficiency (both TE and EE) of each respondent, accommodate exogenous variables that affect inefficiency, and enable testing of statistical hypothesis. The Cobb Douglas functional form was selected because it offers a more parsimonious structure, provides stable estimates for the available data at farm level, allows straightforward interpretation of production elasticities, and reduces the potential statistical issues that may arise from the more complex Translog specification. The dependent variable was organic rice production, while the independent variables include land, seeds, organic fertilizers, organic pesticides, labour, and planting systems. The stochastic frontier production was used to analyze the factors affecting organic rice production, which was transformed into natural logarithm, written as Eq. (1):

$$\ln Y = \beta_0 + \beta_1 \ln X_1 + \beta_2 \ln X_2 + \beta_3 \ln X_3 + \beta_4 \ln X_4 + \beta_5 \ln X_5 + \beta_6 \ln X_6 + \beta_7 \ln X_7 + \beta_8 \ln X_8 + \beta_9 \ln X_9 + \beta_{10} \ln X_{10} + d_1 D_1 + (v_i - u_i) \quad (1)$$

where,

Y = organic rice production (kg);

β_0 = constant;

$\beta_1 - \beta_{10}$ = regression coefficient of each variable;

X_1 = land area (m²);

X_2 = seeds (kg);

X_3 = homemade solid organic fertilizer (kg);

X_4 = manufactured solid organic fertilizer (kg);

X_5 = homemade liquid organic fertilizer (L);

X_6 = manufactured liquid organic fertilizer (L);

X_7 = homemade liquid organic pesticide (L);
 X_8 = manufactured liquid organic pesticide (L);
 X_9 = family labour (workdays);
 X_{10} = non-family labour (workdays);
 d_1 = coefficient of planting system dummy variable;
 D_1 = planting system (dummy variable), (1 = *jajar legowo* planting system, 0 = *tegel* planting system);
 v_i = two-sided random error term (statistical noise);
 u_i = one-sided, non-negative technical inefficiency term ($u_i \geq 0$).

The correlation analysis indicated that the pairwise correlation coefficients among the explanatory variables, homemade solid organic fertilizer, manufactured solid organic fertilizer, homemade liquid organic fertilizer, manufactured liquid organic fertilizer, homemade liquid organic pesticide, and manufactured liquid organic pesticide ranged from -0.199 to 0.032 , which were well below the commonly accepted multicollinearity threshold of ≥ 0.8 . This suggested very weak linear relationships among the inputs; therefore, multicollinearity was not considered a concern in the model. The planting system used was the *jajar legowo* system, which involved alternating several rows of rice plants with a single empty row (alley). This system aims to optimize sunlight exposure, improve air circulation, and facilitate crop management activities such as fertilization and pest control, that has the potential to increase rice yields. Common *jajar legowo* patterns include 2:1, consisting of two planted rows followed by one empty row, with a 20 cm spacing between planted rows and a 40 cm-wide alley, and 4:1, which has four planted rows followed by one empty row. In contrast, the *tegel* planting system maintains uniform spacing of 20×20 cm between rows and plants.

Technical inefficiency is related to various external factors and the socioeconomic characteristics of farmers. The equation for factors of technical inefficiency in organic rice farming in Magelang Regency is described in Eq. (2):

$$U_i = \delta_0 + \delta_1 Z_1 + \delta_2 Z_2 + \delta_3 Z_3 + \delta_4 Z_4 + \delta_5 Z_5 \quad (2)$$

where,

U_i = technical inefficiency term for farmer i ($U_i \geq 0$);
 δ_0 = constant;
 $\delta_1 - \delta_5$ = coefficients of inefficiency determinants;
 Z_1 = farmer age (years);
 Z_2 = education level (score);
 Z_3 = farming experience (years);
 Z_4 = farmer participation in farmer group (score);
 Z_5 = access to agricultural extension and training (score).

The software Frontier 4.1 was used to analyze TE in organic rice farming, based on the calculation set out in Eq. (3):

$$TE_i = \frac{Y_i}{Y_i^*} = \frac{E(Y_i|U_i, X_i)}{E(Y_i|U_i = 0, X_i)} = E[\exp(-U_i)|\varepsilon_i] \quad (3)$$

where,

TE_i = technical efficiency of farmer i ;
 Y_i = production of farmer i ;
 Y_i^* = potential production (obtained from stochastic frontier);
 X_i = actual input vector;
 $\varepsilon_i = v_i - u_i$, composite error term.

The value of TE ranges from zero to one, and this analysis can only be performed using cross-sectional data (Kumbhakar & Wang, 2015). Farmers can be considered technically efficient if their efficiency value is greater than or equal to 0.7 and inefficient if it is less than 0.7 (Alemu et al., 2024; Bhat et al., 2026). The combination of TE and AE will result in EE. EE is the ratio between the minimum total production cost (C_i^*) and the actual total production cost (C_i) (Ogundari & Ojo, 2006), which can be written as Eq. (4) below:

$$EE_i = \frac{C_i^*}{C_i} = \frac{E(C_i|U_i = 0, Y_i, P_i)}{E(C_i|U_i, Y_i, P_i)} = E[\exp(-U_i)|\varepsilon_i] \quad (4)$$

where,

EE_i = economic efficiency (EE values range from 0 to 1);
 C_i^* = minimum total production cost;

C_i = actual total production cost;
 P_i = input price vector for farmer i ;
 $\varepsilon_i = v_i + u_i$, composite error term.

Analysis of the EE of organic rice farming in Magelang Regency was conducted using a stochastic frontier cost function model, as shown in Eq. (5):

$$\ln C_i = \alpha_0 + \alpha_y \ln Y + \alpha_1 \ln P_1 + \alpha_2 \ln P_2 + \alpha_3 \ln P_3 + \alpha_4 \ln P_4 + \alpha_5 \ln P_5 + \alpha_6 \ln P_6 + \alpha_7 \ln P_7 + \alpha_8 \ln P_8 + \alpha_9 \ln P_9 + \alpha_{10} \ln P_{10} + (v_i + u_i) \quad (5)$$

where,

C_i = actual total production cost (Indonesian rupiah, IDR);
 α_0 = constant;
 α_y = coefficient of output;
 Y = organic rice production (kg);
 $\alpha_1 - \alpha_{10}$ = regression coefficient of each variable;
 P_1 = land rent price (IDR/m²);
 P_2 = seed price (IDR/kg);
 P_3 = price of homemade solid organic fertilizer (IDR/kg);
 P_4 = price of manufactured solid organic fertilizer (IDR/kg);
 P_5 = price of homemade liquid organic fertilizer (IDR/L);
 P_6 = price of manufactured liquid organic fertilizer (IDR/L);
 P_7 = price of homemade liquid organic pesticide (IDR/L);
 P_8 = price of manufactured liquid organic pesticide (IDR/L);
 P_9 = family labour wages (IDR/workdays);
 P_{10} = non-family labour wages (IDR/workdays);
 v_i = two-sided random error term (statistical noise);
 u_i = one-sided, non-negative cost inefficiency term ($u_i \geq 0$).

Homemade organic inputs are made from natural ingredients, such as garlic, ginger for pesticide, rice-washing water for fertilizer, and other similar materials, using simple production processes. In contrast, manufactured organic inputs are produced through more modern and standardized processing methods. In terms of cost, homemade inputs tend to be cheaper because they use readily available natural materials and involve simple preparation processes. In addition, the prices of homemade inputs are relatively uniform, while the prices of manufactured inputs vary depending on the product brand.

EE is obtained by multiplying TE and AE. As a result, the value of AE can be determined by Eq. (6) as follows:

$$AE = \frac{EE}{TE} \quad (6)$$

3. Results and Discussion

3.1 Characteristics of Organic Rice Farmers

The 125 organic rice farmers who participated in this study were selected from 22 organic rice farmer groups. The characteristics of the farmers were categorized based on age, level of education, experience in organic rice farming, number of family members, cultivated land area, and land status, as presented in Table 2.

Based on age, the youngest organic rice farmers in Magelang Regency were 27 years old and the oldest were 86 years old. The average age of organic rice farmers was 54 years, which was included in the productive age category (Wang & Zhong, 2026). At a productive age, a person is physically fit and able to perform activities properly and be responsive to changes (Gonzales et al., 2015).

The level of education influenced the adoption of innovation and problem-solving skills (Gusti et al., 2022). The level of education of organic rice farmers in Magelang Regency was quite high, with 70% having a high school or college education. Experience in organic rice farming influenced productivity; farmers with a longer period of experience were more skilled in cultivation and farm management (Hidayati et al., 2019). In this study, the experiences in organic rice farming varied considerably, ranging from 1 to 55 years, with an average of 12 years.

Family members played an important role in farming activities, and their contribution depended on gender, family dynamics, and the scale of the business (Rahman et al., 2020). Most organic rice farmers in Magelang had two to three family members who could help with farming activities, such as plant maintenance, harvesting, and post-harvesting.

Based on land area, most of the land for organic rice farming was less than 4,400 m², with an average of 4,175 m².

The land used for farming included owned land, leased land, and sharecropped land. This situation is in line with organic rice farming land in Kulon Progo Regency, namely owned, leased, and sharecropped land (Istiyanti et al., 2024). Farmers who did not own land could lease it at a cost of IDR 500/m²/season. In addition, farmers could cultivate other people's land and share the yield with the landowner. The tenant farmer bore all production costs and received 50% of the total yield.

Table 2. Characteristics of organic rice farmers

Characteristics	Frequency (Person)	Percentage (%)	Characteristics	Frequency (Person)	Percentage (%)
Age (years)			Farming experience (by year)		
27–38	7	5.6	1–11	78	62.4
39–50	40	32.0	12–22	38	30.4
51–62	54	43.2	23–33	5	4.0
63–74	20	16.0	34–44	2	1.6
75–86	4	3.2	45–55	2	1.6
No. of family members			Land area (m ²)		
0	11	8.8	500–4,400	86	68.8
1	26	20.8	4,401–8,300	27	21.6
2	33	26.4	8,301–12,200	8	6.4
3	45	36.0	12,201–16,100	1	0.8
4	10	8.0	16,101–20,000	3	2.4
Education level			Land status		
Elementary school	20	16.0	Owned	96	76.8
Junior high school	18	14.4	Leased	7	5.6
Senior high school	74	59.2	Sharecropped	22	17.6
College	13	10.4			

3.2 Stochastic Frontier Analysis

SFA was used to measure the TE of farmers in organic rice farming by considering random factors (random error) and inefficiencies (inefficiency term). This model separated output deviations caused by random factors such as climatic conditions and measurement errors from deviations due to technical inefficiencies of farmers (Coelli et al., 2005).

Based on Table 3, the land coefficient was 1.0084 and had a significant effect on organic rice production at a 1% error rate, meaning that if the land was increased by 1%, organic rice production would increase by 1.0084% ceteris paribus. The average land used for organic rice farming was relatively small at 4,175 m², so it needed to be increased by renting other people's land or cultivating other people's land with a sharecropping system. This finding is in line with the results of Noormansyah & Cahrial (2020) and Sulistyowati et al. (2019), who stated that land area was a dominant factor that increased TE because greater economies of scale allowed more optimal use of inputs.

Table 3. Factors that influence organic rice production in Magelang Regency

Variable	Coefficient	Std. Error	Test Statistic	Sig.
Constant	−0.4696	0.6227	−0.7541	ns
Land area	1.0084	0.1104	9.1290	***
Seed	−0.2365	0.1027	−2.3009	**
Homemade solid organic fertilizer	0.0270	0.0218	1.2365	ns
Manufactured solid organic fertilizer	0.0233	0.0214	1.0883	ns
Homemade liquid organic fertilizer	−0.0109	0.0199	−0.5469	ns
Manufactured liquid organic fertilizer	−0.0139	0.0209	−0.6669	ns
Homemade liquid organic pesticide	0.0350	0.0173	2.0137	**
Manufactured liquid organic pesticide	0.0531	0.0233	2.2780	**
Family labour	0.1232	0.0605	2.0347	**
Non-family labour	0.0344	0.0797	0.4316	ns
Planting system	−0.1776	0.0935	−1.8988	*
Sigma-square (σ^2)	0.2212	0.0279	7.9197	***
Gamma (γ)	0.0520	0.3251	0.1600	ns
Likelihood ratio (LR) test of the one-sided error	11.8763			

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$, "ns" indicates not significant ($p \geq 0.10$).

Seed had a negative coefficient (-0.2365) and was significant at a 5% error rate, meaning that an increase in seed use by 1% could reduce organic rice production by 0.2365% *ceteris paribus*. Farmers already used seeds according to the Standard Operating Procedures developed by the GATOS Cooperative, so raising their use would increase competition among plants for nutrients, light, and water, ultimately lowering productivity (Ahmed et al., 2021).

Manufactured solid organic fertilizer had a positive coefficient (0.0233) but was not significant, meaning that if the use of manufactured solid organic fertilizer increased, there was a tendency for organic rice production to increase, with other factors remaining constant. This could be due to the quality of the fertilizer or poor dose of application (Asri et al., 2021).

Homemade liquid organic pesticides had a positive and significant effect at the 5% level with a coefficient of 0.035. This means that a 1% increase in the use of homemade liquid organic pesticides would increase organic rice production by 0.035% of *ceteris paribus*. These results are consistent with the findings of Galluzzo (2023) and Krisdiyanto et al. (2021). The field conditions indicated that the use of homemade liquid organic pesticides was still very low, so it needed an increase to optimize rice crop yields.

Manufactured liquid pesticides had a positive and significant effect at the 5% level with a coefficient of 0.0531, meaning that if the use of manufactured liquid organic fertilizer increased by 1%, organic rice production would increase by 0.0531% *ceteris paribus*. This showed that organic pest control with manufactured liquid pesticides could significantly increase production, in line with the research by Ahmed et al. (2021).

Family labor had a positive and significant effect at the 5% level with a coefficient of 0.1232, meaning that increased involvement of family labor could significantly increase organic rice production. This reflects the important role of family labour in organic farming systems, where careful crop management and intensive supervision were required, in line with research by Selim (2012).

Two planting systems were employed by the organic rice farmers in Magelang, namely *jajar legowo* and *tegel*. The dummy variable of the planting system had a negative and significant effect at the 10% level (coefficient = -0.1776), meaning that organic rice production under the *jajar legowo* planting system was lower than under the *tegel* planting system. This situation differed from research by Istiyanti (2021) which found that the productivity of the *jajar legowo* system was higher than that of the conventional system. The field conditions showed that the farmers applied various *jajar legowo* systems such as 2:1, 3:1, 4:1 and there were pest attacks. These findings are supported by research conducted by Arianti et al. (2022) and Saeri et al. (2021), which discovered that the *jajar legowo* system was likely to reduce rice production. This decline was caused by environmental conditions, seed quality, planting distances that did not follow recommended guidelines, and pest attacks.

The variables of homemade liquid organic fertilizer and manufactured liquid organic fertilizer had negative coefficients but were not significantly influential, meaning that if their use increased, there was a tendency to decrease organic rice production. Homemade solid organic fertilizer and non-family labour had positive but insignificant coefficients, which meant that increasing their use tended to increase organic rice production.

The sigma-squared value (σ^2) of 0.2212 indicated variation in output that could not be explained by production inputs. The estimated gamma parameter was relatively small ($\gamma = 0.052$) and statistically insignificant, indicating that only a minor share of the total variation in organic rice output was attributable to technical inefficiency. According to the stochastic frontier framework, a low gamma value implies that random shocks dominate the residual variance rather than inefficiency effects (Coelli et al., 2005). In organic rice farming, such stochastic influences may arise from agroclimatic variability, biological pest and disease pressures, as well as heterogeneity in the quality of organic input, all of which are largely exogenous to farmers' managerial decisions (Reganold & Wachter, 2016).

3.3 Technical, Allocative, and Economic Efficiency

The likelihood ratio (LR) test value of 11.8763 was significant in the critical χ^2 value at the 10% level (approximately 9.23), indicating that the null hypothesis of no technical inefficiency was rejected. The rejection of the null hypothesis implied that the Ordinary Least Squares (OLS) model and the frontier model were statistically different. This result confirms the existence of technical inefficiency, indicating heterogeneity in TE levels among farmers.

Based on SFA, the TE level of the organic rice farmers ranged from 0.4149 to 0.9875, with an average value of 0.9033 (Table 4). The average TE value indicated that farmers had been able to allocate inputs optimally. There was only a rather small opportunity to increase production from that achieved by the farmers with maximum production levels using the best management system (Zeng & Hu, 2023). With the average production level currently achieved by the farmers, the opportunity to increase maximum production without changing the technology used in organic rice farming was 8.53% ($1 - 0.9033/0.9875$). Based on the distribution of TE levels, 72.8% of farmers were in the high efficiency group (TE = 0.91–1.00).

As many as 91.2% of organic rice farmers in Magelang were technically efficient because their efficiency level was >0.7 (Coelli et al., 2005). The field conditions showed that the farmers used production factors based on the

Standard Operating Procedures developed by members and administrators of the GATOS Cooperative. The seeds used were from organic fields of good quality and were used at a rate of 25 kg/hectare. The fertilizers used were purely organic in the form of manure, solid and liquid organic fertilizers manufactured according to the recommended dosage. The farmers used both homemade pesticides and manufactured pesticides to control pests and diseases in organic rice crops. The farmers engaged in organic rice farming received assistance from agricultural extension workers and leaders of farmer groups.

This high level of efficiency is in line with previous studies (Han et al., 2026; Hidalgo et al., 2025) which stated that organic farmers tended to have higher TE when supported by guidance, training, and good land management. In addition, the high level of TE reflected the success of farmers in implementing more knowledge-intensive organic farming practices, as stated by Wollni & Andersson (2014).

Table 4. Technical efficiency (TE) level of organic rice farming in Magelang Regency

TE Level	Number of Farmers	Percentage (%)
<0.5	1	0.8
0.51–0.60	5	4.0
0.61–0.70	5	4.0
0.71–0.80	8	6.4
0.81–0.90	15	12.0
0.91–1.00	91	72.8
Total	125	100.0
Minimum TE	0.4149	
Maximum TE	0.9875	
Average TE	0.9033	

Table 5. Factors that influence technical inefficiency of organic rice farming in Magelang Regency

Variable	Coefficient	Sig.	Std. Error	t-Value
Constant	6.7429	***	1.6673	4.0440
Age	-1.1392	***	0.3479	-3.2744
Level of education	-0.1726	**	0.0838	-2.0584
Farming experience	-0.1599	*	0.0835	-1.9152
Farmer participation in farmer groups	-0.8482	**	0.3908	-2.1699
Access to agricultural extension and training	-0.0885	ns	0.3283	-0.2696

Note: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$, “ns” indicates not significant ($p \geq 0.10$).

The technical inefficiency model showed that several socioeconomic variables of farmers affected the level of technical inefficiency (Ojo & Baiyegunhi, 2020). These variables included age, education, farming experience, farmer participation in farmer groups, and access to agricultural extension and training.

Based on Table 5, age had a negative coefficient (-1.1392) and was significant at the 1% level, indicating that older farmers tended to be more technically efficient in organic rice farming. This result is not in line with previous studies, which stated that older farmers were often slower to adopt innovations or make technical adjustments, as explained in a study on the adoption of agricultural technology (Uaiene, 2011). The significant effect of age indicated that old farmers in Magelang were still capable of managing production efficiently.

The results of the analysis showed that the level of education had a negative coefficient (-0.1726) and was significant at the 5% level. Education could broaden the knowledge of farmers and improve their ability to understand agricultural technology (Coelli et al., 2005). The fact that education level had insignificant influence on technical inefficiency might indicate that in organic rice farming, increased knowledge and skills could be obtained through non-formal education, such as extension and training, compared to formal education (Ali et al., 2022; Istiyanti et al., 2018).

Farming experience had a significant effect at the 10% level with a negative coefficient (-0.1599), meaning that the more experienced the farmer, the lower the level of inefficiency. This situation is in line with research of Istiyanti & Rahmadynda (2025) that farming experience had a negative influence on the technical inefficiency of organic rice farming in Kulonprogo Regency. The longer the experience of organic rice farming, the more efficient the farming was. Experience increased farmers' ability to allocate inputs, adjust doses of organic fertilizer, manage planting times, control pests biologically, make planting decisions, and consistently implement organic rice farming practices.

The results of this study are in line with various studies showing that experience increased farmers' capacity to adopt technology more efficiently (Bakari et al., 2019; Bifarin et al., 2010). Experience also helped farmers cope with uncertainty in production, especially organic farming, which is more sensitive to the conditions of agroecosystem.

Farmer participation in farmer groups had a negative coefficient (-0.8482) and was significant at the 5% level,

indicating that increased participation reduced technical inefficiency. These results are in line with Reviandy et al. (2021) that farmer groups played an important role in improving farm performance by access to shared knowledge, collective learning, and information exchange, which contributed to better agricultural management practices.

Access to agricultural extension and training had a negative coefficient but was insignificant (-0.0885), indicating that more frequent participation in extension services and training tended to reduce technical inefficiency among farmers. This result is consistent with studies by Amrullah et al. (2025) and Ojo & Baiyegunhi (2020), which showed that participation in agricultural extension programs could increase the knowledge of farmers and help overcome various problems encountered in agricultural activities.

Table 6. Allocative efficiency (AE) level of organic rice farming in Magelang Regency

AE Level	Number of Farmers	Percentage (%)
<0.5	15	12.0
0.51–0.60	14	11.2
0.61–0.70	22	17.6
0.71–0.80	24	19.2
0.81–0.90	27	21.6
0.91–1.00	23	18.4
Total	125	100.0
Minimum AE	0.1527	
Maximum AE	1.0000	
Average AE	0.7352	

AE describes the ability of farmers to allocate input based on prices and costs, so the costs incurred are minimal at a certain level of production. Based on the analysis results (Table 6), the AE values of organic rice farmers in Magelang Regency ranged from 0.1527 to 1, with an average of 0.7352. The distribution of AE levels showed that most farmers were in the moderate efficiency group.

The average AE of 0.7352 indicated that the farmers were generally capable of determining the combination of inputs according to the principle of cost efficiency. Based on the average AE, there was an opportunity for cost savings of around 26.48% if input allocation was carried out at the optimal level. This condition could be influenced by several factors, such as variations in organic input prices, considering that some farmers used inputs that they produced themselves, inaccuracy in determining the dosage of organic fertilizers and biological pesticides, and differences in farmers' abilities to manage organic rice farming (Okello et al., 2019).

These findings are in line with research by Bakari et al. (2019), which stated that AE was often lower than TE, especially in organic farming systems with a variety of cultivation methods and uneven input costs. Moreover, Coelli et al. (2005) explained that low AE generally occurred when farmers faced limitations of information about input prices, capital constraints, and lack of access to extension services on topics related to improving farmers' managerial capabilities in conducting farming operations at efficient costs.

EE is a combination of TE and AE. Mathematically, it is the product of TE and AE. This means that EE reflects farmers' ability to produce output at a minimum cost based on existing technology and input prices. The results of the analysis showed that the EE of organic rice farmers ranged from 0.1494 to 0.9300 (Table 7).

Table 7. Economic efficiency (EE) level of organic rice farming in Magelang Regency

EE Level	Number of Farmers	Percentage (%)
<0.5	24	19.2
0.51–0.60	23	18.4
0.61–0.70	20	16.0
0.71–0.80	25	20.0
0.81–0.90	24	19.2
0.91–1.00	9	7.2
Total	125	100.0
Minimum EE	0.1494	
Maximum EE	0.9300	
Average EE	0.6554	

The average EE level of organic rice farming was 0.6554 which meant that the farmers had only achieved about 65.54% of the optimal level of EE. The low level of EE was mainly influenced by low AE, even though the farmers' TE was relatively high. Although farmers could run organic rice farming technically efficiently, they were not yet able to minimize production costs optimally. The relatively low EE of organic rice farming was mainly due to low AE, meaning that there was an imbalance between the combination of inputs used and their relative prices. In the field, farmers were likely to use excessive inputs such as organic fertilizers and botanical pesticides to meet the

nutritional needs of plants as well as controlling pests and diseases. This also resulted in the use of more labor, which directly increased production costs (Hidayati et al., 2019; Istiyanti et al., 2021). Previous studies have found a similar pattern, namely high TE but low AE in both organic and conventional agriculture (Hidalgo et al., 2025; Setyawati et al., 2024). One of the causes is the difference in farmers' ability to understand cost structures, limitations in business management, and uncertainty of organic input prices, which often fluctuate.

4. Conclusions

This study analyzed the TE, AE, and EE of organic rice farming in Magelang Regency using a stochastic frontier approach. The results indicated relatively high average TE among farmers. However, the component of inefficiency is weak, as reflected by a low and statistically insignificant gamma value; this suggests that output variation is largely driven by random and exogenous factors rather than technical inefficiency. AE is more moderate, resulting in lower EE despite favorable technical performance. This finding implies that effective input use in physical terms does not necessarily translate into cost-minimizing behavior under prevailing price conditions. Socioeconomic characteristics such as age, education, farming experience, and participation in farmer groups are associated with lower estimated inefficiency.

Policy implications should therefore focus less on improving TE and more on strengthening managerial capacity, allocation of input cost, and access to standardized organic inputs. Such interventions are more likely to enhance EE and long-term sustainability, while remaining consistent with the empirical limitations identified in this study.

It should also be acknowledged that the sampling strategy, focused on farmer groups consistently practicing organic rice cultivation, may introduce selection bias and potentially lead to upward-biased estimates of TE. As a result, the findings may overrepresent relatively well-performing farmers. Future studies are therefore encouraged to broaden the sample scope to include more diverse farming conditions and levels of adoption, and to improve the generalizability and robustness of the efficiency estimates.

Author Contributions

Conceptualization, E.I. and A.F.R.; methodology, E.I. and A.F.R.; software, E.I. and A.F.R.; validation, E.I., A., and P.U.; formal analysis, E.I. and A.F.R.; investigation, E.I. and A.; resources, E.I.; data curation, E.I. and A.F.R.; writing original draft preparation, E.I., A., and P.U.; writing review and editing, E.I. and A.F.R.; visualization, E.I.; supervision, A. and P.U.; project administration, E.I. and A.F.R.; funding acquisition, E.I. All authors have read and agreed to the published version of the manuscript.

Funding

This study has been funded by the research grant 2025 from Ministry of Education, Culture, Research and Technology of the Republic of Indonesia.

Informed Consent Statement

Informed consent was obtained from all subjects involved in the study.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Acknowledgements

The authors would like to thank Dr. Nur Rahmawati for her contribution to this research. They also sincerely appreciate the organic rice farmers in Magelang Regency, Central Java, who participated as respondents and generously shared their time, insights, and valuable information.

Conflicts of Interest

The authors declare no conflicts of interest.

References

Ahmed, S., Alam, M. J., Hossain, A., Islam, A. K. M. M., Awan, T. H., Soufan, W., Qahtan, A. A., Okla, M. K., & El Sabagh, A. (2021). Interactive effect of weeding regimes, rice cultivars, and seeding rates influence the

- rice-weed competition under dry direct-seeded condition. *Sustainability*, 13(1), 317. <https://doi.org/10.3390/su13010317>.
- Alemu, F. M., Mengistu, Y. A., & Wassie, A. G. (2024). Factor productivity impacts of climate change and estimating the technical efficiency of cereal crop yields: Evidence from sub-Saharan African countries. *PLoS One*, 19(11), e0310989. <https://doi.org/10.1371/journal.pone.0310989>.
- Ali, S., Murtaza, M., Ahmad, W., Bibi, N., Khan, A., & Khan, J. (2022). Does education and farming experience affect technical efficiency of rice crop growers? Evidence from Khyber Pakhtunkhwa, Pakistan. *Sarhad J. Agric.*, 38(3), 1147–1159. <https://doi.org/10.17582/journal.sja/2022/38.3.1147.1159>.
- Amrullah, E. R., Takeshita, H., & Tokuda, H. (2025). Impact of access to agricultural extension on the adoption of technology and farm income of smallholder farmers in Banten, Indonesia. *J. Agribus. Dev. Emerg. Econ.*, 15(3), 531–547. <https://doi.org/10.1108/jadee-06-2023-0143>.
- Arianti, F. D., Nurwahyuni, E., Minarsih, S., & Amri, A. F. (2022). Growth and yield response of rice based on different planting distances in rainfed field. *E3S Web Conf.*, 361, 04002. <https://doi.org/10.1051/e3sconf/202236104002>.
- Asri, M., Idaryani, & Sahardi. (2021). Effectiveness of solid organic fertilizer (SOF) on lowland rice in Maros, South Sulawesi. *IOP Conf. Ser.: Earth Environ. Sci.*, 911, 012049. <https://doi.org/10.1088/1755-1315/911/1/012049>.
- Bakari, U. M., Maurice, D. C., & Vimtim, M. B. (2019). Analysis of technical efficiency among small-scale rain-fed rice (*Oryza Sativa*) farmers in Adamawa State, Nigeria. *Int. J. Adv. Agric. Sci. Technol.*, 6(7), 34–44.
- Bhat, S. A., Paltasingh, K. R., Mir, A. H., & Hamid, I. (2026). Institutional credit and farm technical efficiency: evidence from a field experiment using stochastic frontier analysis. *Int. J. Rural Manag.* Advance online publication. <https://doi.org/10.1177/097300522614237>.
- Bifarin, J. O., Alimi, T., Baruwa, O. I., & Ajewole, O. C. (2010). Determinant of technical, allocative and economic efficiencies in the plantain (*Musa spp.*) production industry, Ondo State, Nigeria. *Acta Hort.*, 879, 199–209. <https://doi.org/10.17660/ActaHortic.2010.879.18>.
- Coelli, T. J., Prasada Rao, D. S., O'donnell, C. J., & Battese, G. E. (2005). *An Introduction to Efficiency and Productivity Analysis (2nd ed.)*. Springer. <https://doi.org/10.1007/b136381>.
- David, W. & Ardiansyah, A. (2020). The transition toward sustainable organic food systems in Indonesia. *Asia Pac. J. Sustain. Agric. Food Energy*, 8(1 & 2), 23–29.
- Galluzzo, N. (2023). How does eliminating the use of pesticides affect technical efficiency in Italian farms? *Bulgar. J. Agric. Sci.*, 29(1), 14–23.
- Gonzales, E., Matz-Costa, C., & Morrow-Howell, N. (2015). Increasing opportunities for the productive engagement of older adults: A response to population aging. *Gerontologist*, 55(2), 252–261. <https://doi.org/10.1093/geront/gnu176>.
- Gusti, I. M., Gayatri, S., & Prasetyo, A. S. (2022). The affecting of farmer ages, level of education and farm experience of the farming knowledge about Kartu Tani beneficial and method of use in Parakan Distric, Temanggung Regency. *J. Litbang Prov. Jawa Teng.*, 19(2), 209–221. <https://doi.org/10.36762/jurnaljateng.v19i2.926>.
- Han, H., Zeng, H., Jiang, M., & Xiong, J. (2026). Agricultural productive services, stage-specific technical efficiency, and sustainable rice-based food systems: Evidence from Jiangsu, China. *Sustainability*, 18(4), 1744. <https://doi.org/10.3390/su18041744>.
- Hidalgo, H. A., Villano, R. A., Lustre, M. S., & Hidalgo, K. E. S. (2025). Productivity and technical efficiency of organic rice farming in Camarines Sur, Philippines. *Int. J. Adv. Sci. Eng. Inf. Technol.*, 15(1), 223–230. <https://doi.org/10.18517/ijaseit.15.1.19927>.
- Hidayati, B., Yamamoto, N., & Kano, H. (2019). Investigation of production efficiency and socio-economic factors of organic rice in Sumber Ngopoh district, Indonesia. *J. Cent. Eur. Agric.*, 20(2), 748–758. <https://doi.org/10.5513/jcea01/20.2.2143>.
- International Fund for Agricultural Development. (2019). *Indonesia 2000002234: UPLANDS Project Project Design Report July 2019*. <https://www.ifad.org/en/w/corporate-documents/projects-programmes/indonesia-2000002234-uplands-project-project-design-report-july-2019>
- International Fund for Agricultural Development. (2025). *Indonesia 2000002234: UPLANDs Project Supervision Report March 2025*. <https://www.ifad.org/en/w/corporate-documents/projects-programmes/indonesia-2000002234-uplands-project-supervision-report-march-2025>
- Istiyanti, E. (2021). Assessing farmers' decision-making in the implementation of *jajar legowo* planting system in rice farming using a logit model approach in Bantul Regency, Indonesia. *E3S Web Conf.*, 232, 01013. <https://doi.org/10.1051/e3sconf/202123201013>.
- Istiyanti, E., Rahayu, L., & Sriyadi. (2018). Efficiency of organic rice farming in Bantul Regency Special Region of Yogyakarta, Indonesia. *Int. J. Food Res.*, 25, S173–S180.
- Istiyanti, E. & Rahmadynda, A. N. (2025). Efficiency of organic rice farming in Nanggulan District, Kulon Progo Regency, the Special Region of Yogyakarta. *IOP Conf. Ser.: Earth Environ. Sci.*, 1518, 012014.

- <https://doi.org/10.1088/1755-1315/1518/1/012014>.
- Istiyanti, E., Ramli, M. F., & Firmansyah, M. N. (2024). Factors affecting the production risk of organic rice in Kulonprogo Regency, Special Region of Yogyakarta, Indonesia. *E3S Web Conf.*, 595, 01021. <https://doi.org/10.1051/e3sconf/202459501021>.
- Istiyanti, E., Wulandari, R., & Widowati, I. (2021). Technical efficiency of semi organic rice farming in Sleman Regency, Special Region of Yogyakarta. *E3S Web Conf.*, 316, 02047. <https://doi.org/10.1051/e3sconf/202131602047>.
- Krisdiyanto, R., Harisudin, M., & Irianto, H. (2021). Technical efficiency of organic rice farming in Ngawi Regency (The case of the Komunitas Ngawi Organic Center). *IOP Conf. Ser.: Earth Environ. Sci.*, 824, 012103. <https://doi.org/10.1088/1755-1315/824/1/012103>.
- Kumbhakar, S. C. & Wang, H.-J. (2015). Estimation of technical inefficiency in production frontier models using cross-sectional data. In S. Ray, S. Kumbhakar, & P. Dua (Eds.), *Benchmarking for Performance Evaluation* (pp. 1–73). Springer India. https://doi.org/10.1007/978-81-322-2253-8_1.
- Nair, C. M., Salin, K. R., Joseph, J., Aneesh, B., Geethalakshmi, V., & New, M. B. (2014). Organic rice–prawn farming yields 20% higher revenues. *Agron. Sustain. Dev.*, 34(3), 569–581. <https://doi.org/10.1007/s13593-013-0188-z>.
- Noormansyah, Z. & Cahrial, E. (2020). Efficiency of production factors and constraints of organic rice farming at rainfed rice. *IOP Conf. Ser.: Earth Environ. Sci.*, 466, 012027. <https://doi.org/10.1088/1755-1315/466/1/012027>.
- Ogundari, K. & Ojo, S. O. (2006). An examination of technical, economic and allocative efficiency of small farms: The case study of cassava farmers in Osun State of Nigeria. *J. Cent. Eur. Agric.*, 7(3), 423–432.
- Ojo, T. O. & Baiyegunhi, L. J. S. (2020). Impact of climate change adaptation strategies on rice productivity in South-West, Nigeria: An endogeneity corrected stochastic frontier model. *Sci. Total Environ.*, 745, 141151. <https://doi.org/10.1016/j.scitotenv.2020.141151>.
- Okello, D. M., Bonabana-Wabbi, J., & Mugonola, B. (2019). Farm level allocative efficiency of rice production in Gulu and Amuru districts, Northern Uganda. *Agric. Econ.*, 7, 19. <https://doi.org/10.1186/s40100-019-0140-x>.
- Panpluem, N., Mustafa, A., Huang, X., Wang, S., & Yin, C. (2019). Measuring the technical efficiency of certified organic rice producing farms in Yasothon Province: Northeast Thailand. *Sustainability*, 11(24), 6974. <https://doi.org/10.3390/su11246974>.
- Pemerintah Kabupaten Magelang (2018). *Laporan Kinerja Instansi Pemerintah (LKJIP) Pemerintah Kabupaten Magelang Tahun 2017* [Government Agency Performance Report (LKJIP) of Magelang Regency 2017]. <https://ppid.magelangkab.go.id/preview/laporan-kinerja-instansi-pemerintah-lkjp-kab-magelang-tahun-2017-1537833600>
- Phantha, C., Prasara-A, J., Boonkum, P., & Gheewala, S. H. (2021). Social sustainability of conventional and organic rice farming in north-eastern Thailand. *Int. J. Glob. Environ. Issues*, 20(1), 42–59. <https://doi.org/10.1504/ijgenvi.2021.120433>.
- Piadozo, M. E. S., Lantican, F. A., Pabuayon, I. M., Quicoy, A. R., Suyat, A. M., & Maghirang, P. K. B. (2014). Rice farmers' concept and awareness of organic agriculture: Implications for sustainability of Philippine organic agriculture program. *J. Int. Soc. Southeast Asian Agric. Sci.*, 20(2), 142–156.
- Pretty, J. & Bharucha, Z. P. (2014). Sustainable intensification in agricultural systems. *Ann. Bot.*, 114(8), 1571–1596. <https://doi.org/10.1093/aob/mcu205>.
- Rahman, M. W., Palash, M. S., Jahan, H., Jalilov, S.-M., & Mainuddin, M. (2020). An empirical investigation of men's views of women's contribution to farming in Northwest Bangladesh. *Sustainability*, 12(9), 3521. <https://doi.org/10.3390/su12093521>.
- Reganold, J. P. & Wachter, J. M. (2016). Organic agriculture in the twenty-first century. *Nat. Plants*, 2(2), 15221. <https://doi.org/10.1038/nplants.2015.221>.
- Reviandy, O. P., Widiyanto, Rusdiyana, E., Rinanto, Y., & Sudibya. (2021). The role of farmer groups in the development of dryland farming in Ketos Village, Paranggupito Subdistrict, Wonogiri Regency. *IOP Conf. Ser.: Earth Environ. Sci.*, 905, 012127. <https://doi.org/10.1088/1755-1315/905/1/012127>.
- Saeri, M., Lativah, E., Antarlina, S. S., & Arifin, Z. (2021). Technical efficiency analysis of rice farmers in Ngawi District, East Java Province. *IOP Conf. Ser.: Earth Environ. Sci.*, 782, 022007. <https://doi.org/10.1088/1755-1315/782/2/022007>.
- Selim, S. (2012). Labour productivity and rice production in Bangladesh: A stochastic frontier approach. *Appl. Econ.*, 44(5), 641–652. <https://doi.org/10.1080/00036846.2010.515203>.
- Setyawati, I. K., Zainuddin, A., Magfiroh, I. S., Rahman, R. Y., & Suciati, L. P. (2024). Stochastic frontier approach on technical efficiency of rice farming in Jember. In *Proceedings of the 2nd International Conference on Neural Networks and Machine Learning 2023 (ICNNML 2023)* (pp. 184–189). https://doi.org/10.2991/978-94-6463-445-7_19.
- Sujianto, S., Ariningsih, E., Ashari, A., Wulandari, S., Wahyudi, A., & Gunawan, E. (2024). Investigating the

- financial challenges and opportunities of organic rice farming: An empirical long-term analysis of smallholder farmers. *Org. Agr.*, 14, 245–261. <https://doi.org/10.1007/s13165-024-00461-w>.
- Sulistyowati, L., Noor, T. I., Karmana, M. H., & Nugraha, A. (2019). Economics efficiency of share cropping system, evidence from West-Java Indonesia. *Int. J. Innov. Creat. Change*, 10(2), 56–74.
- Uaiene, R. N. (2011). Determinants of agricultural technology adoption in Mozambique. In *10th African Crop Science Conference Proceedings, Maputo, Mozambique, 10–13 October 2011* (p. 375–380).
- Wang, Y. & Zhong, Y. (2026). The impact of agricultural labor aging on wheat production efficiency. *Front. Sustain. Food Syst.*, 10, 1676834. <https://doi.org/10.3389/fsufs.2026.1676834>.
- Willer, H. & Lernoud, J. (2019). *The World of Organic Agriculture: Statistics and Emerging Trends 2019*. Research Institute of Organic Agriculture FiBL and IFOAM–Organics International. https://ciaorganico.net/documypublic/486_2020-organic-world-2019.pdf
- Willer, H., Trávníček, J., Meier, C., & Schlatter, B. (2021). *The World of Organic Agriculture: Statistics and Emerging Trends 2021*. Research Institute of Organic Agriculture FiBL and IFOAM–Organics International. <https://www.fibl.org/fileadmin/documents/shop/1150-organic-world-2021.pdf>
- Wollni, M. & Andersson, C. (2014). Spatial patterns of organic agriculture adoption: Evidence from Honduras. *Ecol. Econ.*, 97, 120–128. <https://doi.org/10.1016/j.ecolecon.2013.11.010>.
- Zeng, F. & Hu, Q. (2023). Measurement of agricultural technical efficiency in China and its influencing factors. *Appl. Ecol. Env. Res.*, 21(5), 4839. https://doi.org/10.15666/aecr/2105_48394851.